

Chapter 41: Clinical Management of the Preeclampsia Syndrome — Revision Notes

Williams Obstetrics 26e, pp. 1840–1892 of the PDF. High-yield numbers are in **bold**. The 11 tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 7 figures are included from the PDF.

Big picture: Delivery is the only cure. Management goals: deliver with the least trauma to mother and fetus, a healthy newborn, and full maternal recovery. Severe preeclampsia and eclampsia are treated the same way: **magnesium sulfate** to prevent or stop seizures, **antihypertensives** for BP $\geq 160/110$ to prevent stroke, then **delivery** (usually by induction). Preeclampsia is also a **marker for lifelong cardiovascular, renal and neurological risk**.

1. Diagnosis and Evaluation

Diagnosis

- More frequent 3rd-trimester visits aid early detection; BP rises may be physiological or pathological.
- Suspected early preeclampsia → **more frequent visits**; outpatient surveillance continues unless **overt hypertension, proteinuria, headache, visual changes or epigastric pain** develop.
- Parkland: **new-onset BP ≥ 140 systolic or ≥ 90 diastolic** → **admit** to exclude preeclampsia or define its severity.

Evaluation on admission

- **Daily** check for headache, visual changes, epigastric pain; **daily weight** (rapid gain).
- **Proteinuria** quantification or urine protein:creatinine ratio.
- **BP every 4 h** with an appropriate cuff (more often if previously high).
- **Creatinine, transaminases, CBC with platelets**; frequency by severity. Uric acid, LDH and coagulation tests are of **questioned value**.
- **Fetal size, well-being and amniotic fluid volume**.
- Reduced activity may help (weak evidence). **PIGF and sFit-1** are still investigational.
- Signs rarely resolve fully **until after delivery**.

2. Consideration for Delivery

Objectives

1. End the pregnancy with the **least trauma** to mother and fetus.
2. Birth of a **healthy newborn** who thrives.
3. **Complete restoration** of maternal health.
 - **Termination of pregnancy is the only known cure.**
 - **Headache, visual changes or epigastric pain** → convulsions may be imminent. **Oliguria** is another ominous sign.
 - **Severe preeclampsia = anticonvulsant + antihypertensive → delivery.** Eclampsia is treated identically, **even with an unfavorable cervix**.

Route

- **Labor induction** (usually with cervical ripening); **cesarean for other obstetrical indications** only.
- Parkland (neonates **750–1500 g**, severe preeclampsia): induction in half, **vaginal delivery in 1/3** of those induced. Consortium on Safe Labor: half induced, **half of those delivered vaginally**.

Timing

Situation	Recommendation / data
Preeclampsia without severe features	Deliver after 37 wk (RCT of 756; ACOG). Parkland: induce at 38 wk
Antenatal testing (ACOG)	Twice weekly for gestational hypertension/nonsevere; daily with severe features
HYPITAT-II (nonsevere, 34–37 wk)	Immediate delivery: adverse maternal outcome 1.1% vs 3.1% , but neonatal RDS 5.7% vs 1.7%
Chappell 2019 (901, nonsevere preeclampsia 34 to <37 wk)	Early delivery: fewer severe maternal features , but more perinatal deaths/NICU admissions
PEACOCK	Angiogenic biomarkers did not help decide on late-preterm delivery

- With a preterm fetus, delay is justified in milder cases, with **NST or BPP** surveillance.

3. Inpatient or Outpatient Care (mild–moderate, stable hypertension)

- Reduced activity seems reasonable, but **complete bed rest is NOT recommended** (Task Force 2013); **SMFM: do not prescribe activity restriction**. Bed rest is impractical and predisposes to **thromboembolism**.
- **Parkland antepartum unit** (since 1973, >10,000 nulliparas):
 - Most improve, but **~90% have recurrent hypertension** before or during labor.
 - Modest nursing, **no drugs except iron and folate**, few tests; costs minimal vs NICU; thromboembolism rare.
- **Home care** (1182 nulliparas with mild gestational hypertension, 20% proteinuric; enrolled 32–33 wk, delivered 36–37 wk): severe preeclampsia **~20%**, eclampsia in **2**, HELLP **3%**, FGR **~20%**, PMR **4.2/1000**.
- Outpatient care is fine if disease does not worsen and fetal jeopardy is not suspected; not feasible with **vulnerable home situations**. Home BP/urine protein or visiting nurse helps.
- **Bottom line:** inpatient or close outpatient care is appropriate for **mild de novo hypertension**, including **nonsevere preeclampsia**. Keys: close surveillance, home support, access to care.

Table 41-1. Randomized Clinical Trials Comparing Hospitalization Versus Routine Care for Women with Mild Gestational Hypertension or Preeclampsia

Study Groups	No.	Para ₀ (%)	Chronic HTN (%)	EGA adm (wk)	Prot (%)	EGA del (wk)	<37 wk (%)	<34 wk (%)	Mean Hosp (d)	Mean BW (g)	SGA (%)	PMR (%)
Crowther (1992)	218 ^a											
Hospitalization	110	13	14	35.3	0	38.3	12	1.8	22.2	3080	14	0
Outpatient	108	13	17	34.6	0	38.2	22	3.7	6.5	3060	14	0
Turnbull (2004)	374 ^b											
Hospitalization	125	63	0	35.9	22	39	—	—	8.5	3330	3.8	0
Day Unit	249	62	0	36.2	22	39.7	—	—	7.2	3300	2.3	0

Columns "adm" = maternal characteristics at admission; "del" = at delivery. ^aExcluded women with proteinuria at study entry. ^bIncluded women with ≤1+ proteinuria. BW = birthweight; EGA = estimated gestational age; Hosp =

hospital; HTN = hypertension; Para₀ = nulliparas; PMR = perinatal mortality rate; Prot = proteinuria; SGA = small for gestational age.

- Crowther: outpatients had **~2× severe hypertension and preterm delivery** (<34 and <37 wk), but other outcomes were similar.
- Turnbull (day care, popular in Europe): no perinatal deaths, no eclampsia or HELLP; similar cost; **satisfaction favored day care**.

4. Antihypertensive Therapy for Mild to Moderate Hypertension

- **Disappointing** for early mild preeclampsia/gestational hypertension: the only benefit is **fewer episodes of severe hypertension** (also 49-trial review).
- Labetalol (Sibai 1987a): lower BP but no pregnancy prolongation or outcome gain; **FGR doubled (19% vs 9%)**.
- Chronic hypertension treatment: see Chapter 53.

Table 41-2. Randomized Placebo-Controlled Trials of Antihypertensive Therapy for Early Mild Gestational Hypertension

Study	Study Drug (No.)	Pregnancy Prolonged (d)	Severe HTN ^a (%)	Cesarean Delivery (%)	Placental Abruptio (%)	Mean Birthweight (g)	Growth Restriction (%)	Neonatal Deaths (No.)
Sibai (1987a) ^a , 200 inpatients	Labetalol (100)	21.3	5	36	2	2205	19 ^c	1
	Placebo (100)	20.1	15 ^c	32	0	2260	9	0
Sibai (1992) ^b , 200 outpatients	Nifedipine (100)	22.3	9	43	3	2405	8	0
	Placebo (100)	22.5	18 ^c	35	2	2510	4	0
Pickles (1992), 144 outpatients	Labetalol (70)	26.6	9	24	NS	NS	NS	NS
	Placebo (74)	23.1	10	26	NS	NS	NS	NS
Wide-Swensson (1995), 111 outpatients	Isradipine (54)	23.1	22	26	NS	NS	NS	0
	Placebo (57)	29.8	29	19	NS	NS	NS	0

^aAll women had preeclampsia. ^bIncludes postpartum hypertension. ^c $p < 0.05$ when study drug compared with placebo. HTN = hypertension; NS = not stated.

The printed footnote c reads " $p < 0.5$ ", an obvious typo for $p < 0.05$.

5. Expectant Management of Preterm Severe Preeclampsia

- Aim: better neonatal outcome **without compromising maternal safety**; always with daily (or more frequent) **inpatient** maternal and fetal monitoring. Controversial and **potentially dangerous**.

Key studies

- **Memphis, 18–27 wk (60 women): disastrous.** PMR **87%**; abruption 13, eclampsia 10, renal failure 3, hypertensive encephalopathy 2, ICH 1, ruptured hepatic hematoma 1 (no maternal deaths).
- **Sibai 1994 RCT, 28–32 wk (95 women, HELLP excluded):** aggressive = steroids then delivery in **48 h**; expectant = bed rest + oral labetalol or nifedipine. Pregnancy prolonged **15.4 days**; neonatal outcomes improved.
- At **23–34 wk**: abruption, HELLP, pulmonary edema, renal failure, eclampsia; **PMR averages ~90/1000. FGR common (94%** in Dutch studies), and PMR is disproportionately high in FGR neonates.
- Hence **HELLP and FGR are usually excluded.** (One study suggests "stable" HELLP <34 wk may benefit.)
- **MEXPRE (267, 28–32 wk):** PMR **~90/1000 in both arms**; no neonatal benefit. Expectant arm: **FGR 22% vs 9%, abruption 7.6% vs 1.5%.**
- **Cochrane (6 RCTs, 24–34 wk):** may benefit perinatal outcome; **insufficient data on maternal health.**

Table 41-3. Maternal and Perinatal Outcomes with Expectant Management of Severe Preeclampsia from 23 to 34 Weeks

Study	No.	Days Gained	Placental Abruption (%)	HELLP (%)	Pulm. Edema (%)	AKI (%)	Eclampsia (%)	FGR (%)	PMR (%)
Ganzevoort (2005a,b)	216	11	1.8	18	3.6	NS	1.8	94	18
Bombrys (2009)	66	5	11	8	9	3	0	27	1.5
Abdel-Hady (2010)	211	12	3.3	7.6	0.9	6.6	0.9	NS	48
Vigil-De Gracia (2013)	131	10.3	7.6	14	1.5	4.5	0.8	22	8.7
Mooney (2016)	108	5 ^b	NS	11	NS	NS	2	~36	4.6
Hoshino (2019)	79	8.5	12	12	4	3	3	NS	0
Cavaignac-Vitalis (2019) ^a	87	6 ^b	NS	100	0	NS	3.5	NS	9.3
Total	892								
Range		5–12	1.8–12	7.6–100	0.9–4.0	3–6.6	0.9–18	27–94	1.5–48

Maternal outcomes = abruption through eclampsia; perinatal outcomes = FGR and PMR. ^aAll had HELLP syndrome at enrollment. ^bMedian. AKI = acute kidney injury; FGR = fetal-growth restriction; HELLP = hemolysis, elevated liver enzyme levels, low platelet count; NS = not stated; PMR = perinatal mortality rate; Pulm. = pulmonary.

The printed "Range" row does not match the rows above it. From the rows themselves: pulmonary edema 0–9, eclampsia 0–3.5, FGR 22–94, PMR 0–48.

6. Expectant Management of Midtrimester Preeclampsia (<28 wk)

- Review of 8 studies (~200 women, onset **<26 wk**): maternal complications common.
- **No neonates survived delivery <23 wk** → most experts recommend **termination**.

Gestational age	Perinatal survival
23 wk	18% (long-term morbidity unknown)
24–26 wk	~60%
26 wk	~90%

- Maternal complications (especially **HELLP**) are common and approach **50%**; one mother died; **PMR >50%**. No comparative studies show a perinatal benefit.

Table 41-4. Pregnancy Outcomes with Expectant Management of Women with Severe Midtrimester Preeclampsia^a

Study	No.	EGA	Prolonged Pregnancy	HELLP (%)	Pulm. Edema (%)	Placental Abrupton (%)	Eclampsia (%)	Death (%)	Perinatal Mortality
Gaugler-Senden (2006)	26	<26 wk	24 d ^b	16	4	0	5	1	23/28
Budden (2006)	31	<25 wk	~12 d ^b	20	0	0	0	0	22/31
Bombrys (2008)	46	<27 wk	6 d ^c	11	2	6	1	NS	22/51
Abdel-Hady (2010)	61	24–28 wk	12 d ^b	6	0	4	0	0	29/61
Belghiti (2011)	51	<26 wk	7 d ^c	5	2	3	0	0	31/53

HELLP through Death are maternal outcomes (%). ^aIncludes women with superimposed preeclampsia. ^bMean. ^cMedian. EGA = estimated gestational age; NS = not stated; Pulm. = pulmonary edema.

Corticosteroids

- **For lung maturation:** do not seem to worsen maternal hypertension; ↓ RDS and ↑ survival. Only one RCT (Amorim, **218 women, 26–34 wk**, betamethasone vs placebo): ↓ RDS, IVH and death, **but 2 maternal deaths and 18 stillbirths**.
- **To improve HELLP: NOT recommended** (Task Force 2013). Three RCTs showed no benefit:
 - Dexamethasone vs placebo, 132 women: no difference in stay, lab recovery or complications.
 - Postpartum dexamethasone, 105 women: no advantage (Fig 41-1).
 - Methylprednisolone for platelets 50,000–150,000/μL: no benefit.

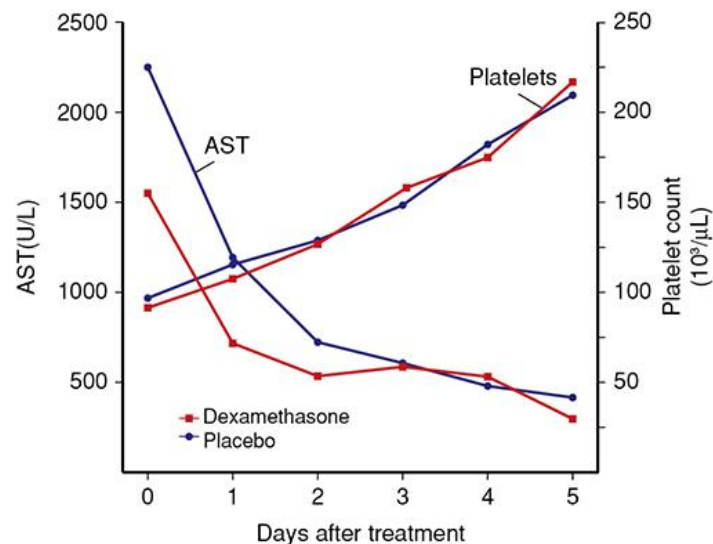


Figure 41-1. Postpartum HELLP: AST falls and platelets recover over ~5 days at the same pace with dexamethasone or placebo.

Expectant management recommendations

- Benefits at **24–32 wk** are **not overwhelming** compared with the maternal risks.
- **SMFM (2011) and the Task Force:** expectant management is a **reasonable alternative in selected women with severe preeclampsia <34 wk**, with inpatient surveillance and strict delivery triggers.
- Vaginal delivery is attempted, but **cesarean becomes more likely at lower gestational ages**.

- **Williams (Parkland) view is more conservative:** maternal safety comes first, and the average gain is **~1 week** without convincing perinatal benefit. If expectant management is used, the Table 41-5 triggers must be **strictly heeded**.

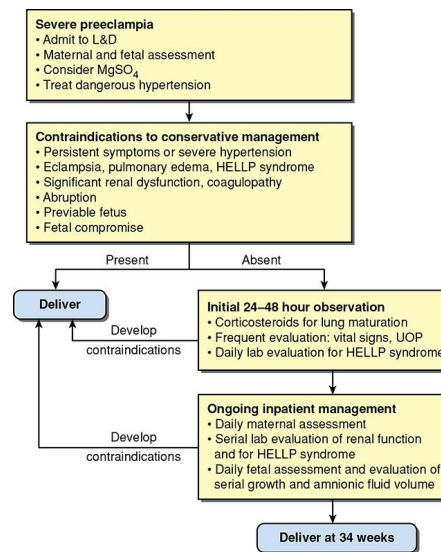


Figure 41-2. Severe preeclampsia <34 wk: admit, MgSO₄, treat BP → deliver if any contraindication; otherwise steroids, 24–48 h observation, daily inpatient surveillance, deliver at 34 wk.

Table 41-5. Indications for Delivery in Women <34 Weeks' Gestation Managed Expectantly

Prompt Delivery After Maternal Stabilization and After Single-dose Corticosteroid Therapy for Lung Maturation ^a	Delay Delivery 48 hr If Possible to Allow Corticosteroid Therapy for Lung Maturation
Uncontrolled severe hypertension	Preterm ruptured membranes or labor
Persistent headaches, refractory to treatment	Fetal-growth restriction
Persistent epigastric pain	Oligohydramnios
Eclampsia	Reversed end-diastolic Doppler flow in umbilical artery
HELLP syndrome	Worsening renal dysfunction
Pulmonary edema	
Placental abruption	
Disseminated intravascular coagulation	
Stroke	
Myocardial infarction	
Nonreassuring fetal status	
Fetal demise	

^aInitial dose only, do not delay delivery. HELLP = hemolysis, elevated liver enzyme levels, low platelet count. From American College of Obstetricians and Gynecologists, 2020b; Society for Maternal-Fetal Medicine, 2011.

Experimental therapies

- **Apheresis** to lower sFlt-1, **RNA silencing** of placental sFlt-1, **esomeprazole**, **sildenafil**, **recombinant antithrombin**. **None has shown promise.**

7. Severe Preeclampsia and Eclampsia

- **Severe features:** SBP **≥160** or DBP **≥110**; transaminitis; thrombocytopenia; renal insufficiency; pulmonary edema; new-onset headache; visual changes (Table 40-2).

- **Eclampsia** = new-onset **tonic-clonic, focal or multifocal seizures** with no other cause.

Eclampsia: maternal complications (399 women, Mattar & Sibai)

Complication	Rate
Placental abruption	10%
Neurological deficits	7%
Aspiration pneumonia	7%
Pulmonary edema	5%
Cardiopulmonary arrest	4%
Acute kidney injury	4%
Maternal death	1%

- Also HELLP, pulmonary embolism, stroke.
- **Preeclampsia almost always precedes convulsions** (sometimes unnoticed).
- Most common in the **3rd trimester**, increasingly so toward term. Postpartum eclampsia has declined (prenatal care, earlier detection, **prophylactic MgSO₄**).
- ⚠ Consider **other diagnoses** if seizures occur **>48 h postpartum**, or with **focal deficits, prolonged coma or atypical** features.

Clinical findings

- **Protect the woman and her airway.** Violent tonic-clonic movements can throw her from bed and cause **tongue biting**; this phase lasts **~1 min**.
- Then postictal state or coma; consciousness usually returns between infrequent seizures, often with a **semiconscious combative** state.
 - Coma persisting between frequent seizures → death may follow. Rare **status epilepticus** needs deep sedation or general anesthesia.
- **Respiratory rate** rises, may reach **≥50/min** (hypercarbia, lactic acidemia, transient hypoxia); cyanosis in severe cases.
- ⚠ **High fever is a grave sign** → likely cerebrovascular hemorrhage.
- Proteinuria usually present. **¼ of severe preeclampsia has some AKI**; oliguria or occasionally anuria. **Hemoglobinuria** may be seen; hemoglobinemia is rare. Edema may be marked.



Figure 41-3. Tongue hematoma from biting during an eclamptic convulsion (thrombocytopenia may add to bleeding).



Figure 41-4. Severe pitting edema in antepartum preeclampsia: fingerprint indentations on both shins.

Recovery after delivery

- **Rising urine output** = an early sign of improvement.
- Proteinuria and edema usually gone **within 1 week**; BP normal in **a few days to 2 weeks**. Persisting severe hypertension suggests **underlying chronic vascular disease**.

Labor and the fetus

- Antepartum eclampsia: labor may start soon after and progress **rapidly**; intrapartum seizures can intensify and shorten labor.
- **Fetal bradycardia** often follows a seizure (maternal hypoxemia, lactic acidemia). It usually recovers in **2–10 min**.
- ⚠ **Bradycardia >10 min** → **think of another cause, e.g., placental abruption**.

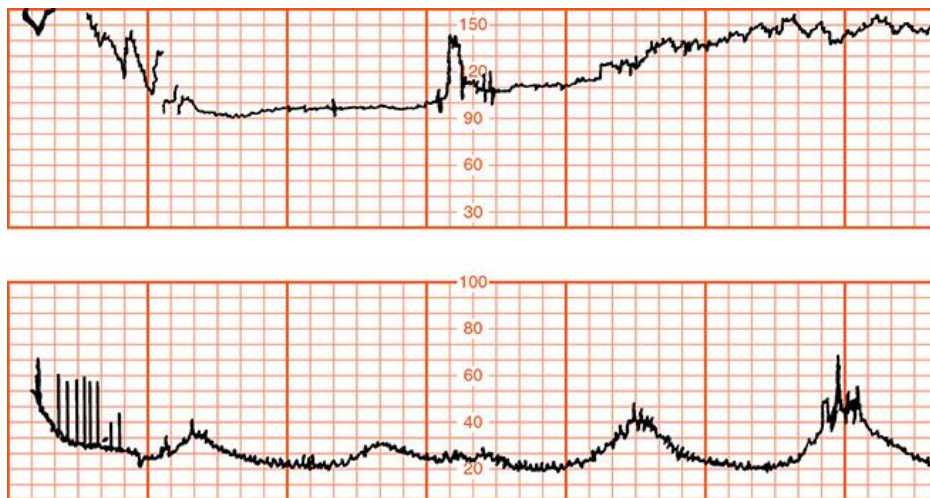


Figure 41-5. Fetal bradycardia after an intrapartum eclamptic seizure, recovering with return of variability in ~5 min. Don't rush to cesarean.

Serious sequelae

- **Pulmonary edema**: right after the seizure or hours later.
- **Sudden death** with a seizure → usually **massive cerebral hemorrhage**. Sublethal hemorrhage → **hemiplegia**. More likely in **older women with chronic hypertension**.
- New **neurological deficit** → **emergent cranial CT**.
- **Altered consciousness/persistent coma in up to 5%** (cerebral edema; herniation may kill).
- **Blindness in ~10%** after a seizure; usually improves.
- **Postictal psychosis** (rare): lasts days to 2 weeks; good prognosis; antipsychotics effective.

Differential diagnosis

- Epilepsy, encephalitis, meningitis, brain tumor, **neurocysticercosis**, **amniotic fluid embolism**, **postdural puncture headache**, **ruptured cerebral aneurysm**.
- Eclampsia is **overdiagnosed more often than missed**, but **every pregnant woman with convulsions has eclampsia until proven otherwise** → **load with MgSO₄** while other diagnoses are explored.

8. Management of Severe Preeclampsia–Eclampsia

US philosophy (four principles)

1. **Control or prevent convulsions** with an **IV loading dose of MgSO₄**, then maintenance (usually IV).
2. **Intermittent antihypertensives** for dangerously high BP.
3. **Avoid diuretics** (unless obvious pulmonary edema), **limit IV fluids** (unless excessive loss), **avoid hyperosmotic agents**.
4. **Deliver** to resolve preeclampsia.

Magnesium sulfate

- Effective **anticonvulsant without CNS depression**. Given **IV infusion** (almost universal in the US), **IM** (equally effective; useful where infusion is not available), or intermittent **IV 2-g** injections.
- Same doses for **severe preeclampsia and eclampsia**. Given **during labor and for 24 h postpartum** (labor and delivery is when seizures are most likely).
- ⚠ **MgSO₄ is NOT an antihypertensive**. It acts on the **cerebral cortex**.
- Usually seizures stop after the **4-g load**; orientation returns in **1–2 h**.

Table 41-6. Magnesium Sulfate Dosage Schedule for Severe Preeclampsia and Eclampsia

Continuous Intravenous (IV) Infusion

Give **4- to 6-g loading dose** of magnesium sulfate diluted in 100 mL of IV fluid administered over **15–20 min**

Begin **2 g/hr** in 100 mL of IV maintenance infusion. Some recommend 1 g/hr

Monitor for magnesium toxicity: assess **deep tendon reflexes** periodically; some measure serum magnesium level at 4–6 hr and adjust infusion to maintain levels between **4 and 7 mEq/L (4.8–8.4 mg/dL)**; measure serum magnesium levels if serum **creatinine ≥1.0 mg/dL**

Magnesium sulfate is discontinued **24 hr after delivery**

Intermittent Intramuscular Injections

Give **4 g** of magnesium sulfate (MgSO₄·7H₂O USP) as a **20% solution IV** at a rate not to exceed **1 g/min**

Follow promptly with **10 g of 50%** magnesium sulfate solution, one half (**5 g**) injected deeply in the upper outer quadrant of **each buttock** through a 3-inch-long 20-gauge needle. (Addition of 1.0 mL of 2% lidocaine minimizes discomfort.) If convulsions persist after 15 min, give up to **2 g more IV** as a 20% solution at a rate not to exceed 1 g/min. If the woman is large, up to 4 g may be given slowly

Every 4 hr thereafter, give **5 g of 50%** solution injected deeply in the upper outer quadrant of alternate buttocks, but only after ensuring that: the **patellar reflex is present**, **respirations are not depressed**, and **urine output the previous 4 hr exceeded 100 mL**

Magnesium sulfate is discontinued **24 hr after delivery**

Memory hook (Pritchard IM regimen): "4 IV + 10 IM (5 each buttock), then 5 IM q4h — only if reflexes, respirations and urine (>100 mL/4 h) are all present."

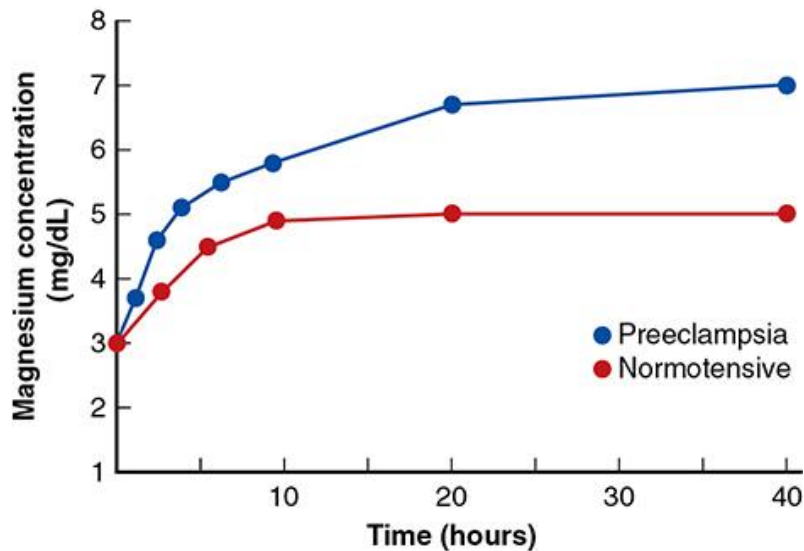


Figure 41-6. Serum Mg after a 4-g load + 2 g/h: preeclamptic women reach ~7 mg/dL versus ~5 mg/dL in normotensive women.

Recurrent seizures on MgSO₄

- **Up to 15%** convulse again → give **another 2 g of 20% MgSO₄ IV slowly** (once in a small woman, up to twice in a larger woman).
- Only **5 of 245** eclamptic women at Parkland needed another anticonvulsant → small IV dose of **midazolam or lorazepam**. Avoid prolonged benzodiazepine use (↑ **mortality from aspiration pneumonia**).

Duration

- Traditionally **24 h after delivery**; for postpartum eclampsia, **24 h after seizure onset**. Shorter courses (12 h, or stopping at delivery) need more study.

Pharmacology and toxicology

- Cleared **almost entirely by the kidney**; Mg clearance $\approx \frac{1}{3}$ of GFR. Toxicity is unusual when GFR is normal or only slightly reduced.
- Mg excretion is **not urine-flow dependent** → urine volume alone does not show renal function; **measure serum creatinine**.
- **Load normally regardless of renal function** (4–6 g); **only the maintenance rate is reduced** when GFR is low. With creatinine **>1.0 mg/dL**, check serial serum Mg levels to guide the infusion.
- **Obesity: >60%** of women with BMI >30 on 2 g/h were **subtherapeutic at 4 h** → most obese women need **3 g/h**. (1 g/h gives low levels.) Routine Mg levels are still **not generally recommended**.

The chapter text prints creatinine ">1.0 mg/mL"; Table 41-6 gives the correct unit, mg/dL.

Plasma magnesium	Effect
4–7 mEq/L (4.8–8.4 mg/dL; 2.0–3.5 mmol/L)	Therapeutic: prevents/arrests eclamptic seizures
8–10 mEq/L	Needed to inhibit uterine contractions
10 mEq/L (~12 mg/dL)	Patellar reflexes lost (curariform action) = warning sign
>10 mEq/L	Breathing weakens
≥12 mEq/L	Respiratory paralysis and arrest

Treating toxicity

- **Stop MgSO₄ + calcium gluconate or calcium chloride 1 g IV** → reverses mild–moderate respiratory depression (keep it at the bedside). Effect may be short-lived at a steady-state toxic level.
- Severe depression or arrest → **prompt intubation and ventilation**.
- Direct myocardial toxicity is uncommon.

Cardiovascular and other effects

- 4 g IV over 15 min → MAP falls slightly, **cardiac index ↑ 13%** (↓ SVR). Transient nausea and flushing; effect lasts only **15 min**. Weaker vasodilation in placental vessels.
- Small rise in **CSF magnesium**, proportional to serum level.
- Anticonvulsant/neuroprotective mechanisms: ↓ presynaptic **glutamate** release, **NMDA receptor blockade**, potentiation of adenosine, better mitochondrial calcium buffering, block of voltage-gated calcium entry.
- **Uterus**: needs **8–10 mEq/L** to inhibit contractions; standard regimens cause only transient depression after the load. **Delivery blood loss is not increased**.

Neuroprophylaxis: seizure prevention

- In all RCTs **MgSO₄ beat the comparator**.
- **Magpie** (>10,000 women with severe preeclampsia, 33 countries): **58% ↓ eclampsia** vs placebo; ↓ maternal mortality and abortion; child behavior at 18 months unchanged.
- **Nimodipine** (cerebral vasodilator CCB): eclampsia **>3×** higher than with MgSO₄ (**2.6% vs 0.8%**).

Table 41-7. Randomized Comparative Trials of Prophylaxis with Magnesium Sulfate and Placebo or Another Anticonvulsant in Women with Gestational Hypertension

Study / Inclusions	Magnesium Sulfate: Seizures/Treated (%)	Control: Seizures/Treated (%)	Comparison ^a
Lucas (1995): gestational hypertension ^b	0/1049	Phenytoin 10/1089 (0.9)	p <0.001
Coetsee (1998): severe preeclampsia	1/345 (0.3)	Placebo 11/340 (3.2)	RR = 0.09 (0.1–0.69)
Altman (2002) ^c : severe preeclampsia	40/5055 (0.8)	Placebo 96/5055 (1.9)	RR = 0.42 (0.26–0.60)
Belfort (2003): severe preeclampsia	7/831 (0.8)	Nimodipine 21/819 (2.6)	RR = 0.33 (0.14–0.77)

^aAll comparisons significant p <0.05. ^bIncluded women with and without proteinuria and those with all severities of preeclampsia. ^cMagpie Trial Collaboration Group, 2002. RR = relative risk.

The Coetsee lower confidence limit prints as 0.1, which is above the RR of 0.09. It is probably 0.01.

Who should get MgSO₄?

- It prevents proportionately more seizures in **worse disease**, but severity is hard to quantify.
- **ACOG: give to all women with eclampsia or severe preeclampsia**.
- Nonsevere disease without MgSO₄: **~1 in 100 seizes**, and **¼ of those need emergent cesarean** (general anesthesia risk). ACOG accepts **not treating 99 to risk eclampsia in 1**.
- **Parkland policy**: MgSO₄ for preeclampsia **with severe features**, and conservatively for **proteinuric hypertension**.

Fetal and neonatal effects

- Crosses the placenta rapidly (equilibrates in fetal serum, less in amniotic fluid).
- FHR: small but significant ↓ **variability**, lower (still normal) baseline, fewer prolonged decelerations.
- Generally **safe**; need for resuscitation not related to cord Mg (>1500 preterm neonates).
- Parkland (6654 mostly term newborns): **hypotonia 6%**, lower 1- and 5-min Apgars, more intubation and special-care admissions. **Neonatal depression only with severe hypermagnesemia**.
- **Protects against cerebral palsy in VLBW** newborns (see Ch 45); insufficient data at term.

Maternal safety and efficacy: Eclampsia Trial Collaborative Group (1687 eclamptic women)

- MgSO₄ had significantly fewer **recurrent seizures** (aggregate **9.7%**) and fewer **maternal deaths (3.2% vs 5.2%)** than diazepam or phenytoin.

Table 41-8. Randomized Comparative Trials of Magnesium Sulfate versus Phenytoin and Diazepam to Prevent Recurrent Eclamptic Convulsions

	Magnesium Sulfate	Phenytoin	Diazepam
Recurrent seizures ^a	60/453 (13%)	—	126/452 (28%)
	22/388 (5.6%)	66/389 (17%)	—
Maternal deaths ^b	10/388 (2.6%)	20/387 (5.2%)	—
	17/453 (3.8%)	—	24/452 (5.3%)

^aAll comparisons $p < 0.01$. ^bIndividual comparisons nonsignificant, combined comparison $p < 0.05$.

"17.453" in the printed table is a typo for 17/453. The printed table (and the chapter text) also put the phenytoin and diazepam labels the wrong way round in the seizure rows. Each arm has a fixed size: 453 vs 452 women in the diazepam comparison and 388 vs 387 in the phenytoin comparison, as the deaths rows show. So the seizure rows are corrected here: recurrent seizures 28% with diazepam and 17% with phenytoin (Collaborative Eclampsia Trial, 1995). Either way, $MgSO_4$ is superior to both.

Severe hypertension treatment

- Dangerous BP → **cerebral hemorrhage**, hypertensive encephalopathy, eclampsia, **abruption**, heart failure. Treat **systolic >160**.
- **Target (NHBPEP, ACOG): SBP ≤ 160 and DBP ≤ 110 .** ⚠ Lower diastolic pressures can **compromise placental perfusion**.
- **At least half** of serious hemorrhagic strokes with preeclampsia occur in women with **chronic hypertension**: long-standing hypertension → **Charcot-Bouchard aneurysms** in lenticulostriate branches of the MCA (basal ganglia, putamen, thalamus, deep white matter, pons, deep cerebellum), which rupture with sudden BP surges.
- **First-line (ACOG): hydralazine, labetalol, nifedipine.** All **equally effective** and reduce stroke risk.
 - Hydralazine → more **tachycardia and palpitations**.
 - Labetalol → more **hypotension and bradycardia**.
 - Nifedipine lowers BP **faster** than labetalol. Both reduce FHR accelerations.
- Others tested but not widely used: verapamil, nitroglycerin, nitroprusside, ketanserin, nicardipine, nimodipine.

Table 41-9. Antihypertensive Agents for Urgent Control of Severe Hypertension

Drug	Dose	Side Effects
Hydralazine	5 mg IV or IM followed by 10-mg doses at 15–20 min intervals for up to 3 doses or constant infusion 0.5–10 mg/hr	Tachycardia, headaches, hypotension
Labetalol	10 mg IV followed by 20 mg IV in 20 min , then 40 mg IV , then 80 mg IV or constant infusion 1–2 mg/min	Asthma precipitation, bradycardia, hypotension
Nifedipine	10 mg orally followed by 20 mg doses at 20 min × 2	Tachycardia, headaches

IM = intramuscular; IV = intravenous.

The table says the second labetalol dose comes "in 20 min", but the text gives 10-minute steps (10 → 20 → 40 → 40 → 80 mg).

Details from the text

Drug	Parkland protocol	ACOG
Hydralazine	Always start with 5 mg (response is unpredictable, regardless of BP height); acts in ~10 min; then 10 mg every 15–20 min; Parkland will give a 3rd dose	Switch to labetalol if still severe after the 2nd dose
Labetalol (α + nonselective β -blocker)	10 → 20 → 40 → 40 → 80 mg IV at 10-min intervals. ⚠️ Avoid in asthma	10–20 mg IV, then 20–80 mg every 10–30 min; then hydralazine
Nifedipine (immediate release)	10–20 mg orally, repeat 10–20 mg after 20 min; then labetalol. ⚠️ Never sublingual (dangerously rapid effect)	—

- ⚠️ **Overtreatment:** avoid a large initial hydralazine dose. Fig 41-7: hydralazine every 5 min instead of every 15 → MAP **180 → 90 mm Hg** within 1 h (BP 270/150 → 110/80) with **fetal bradycardia/decelerations**, reversed by rapid crystalloid. This can be mistaken for **abruption** → unnecessary emergency cesarean.

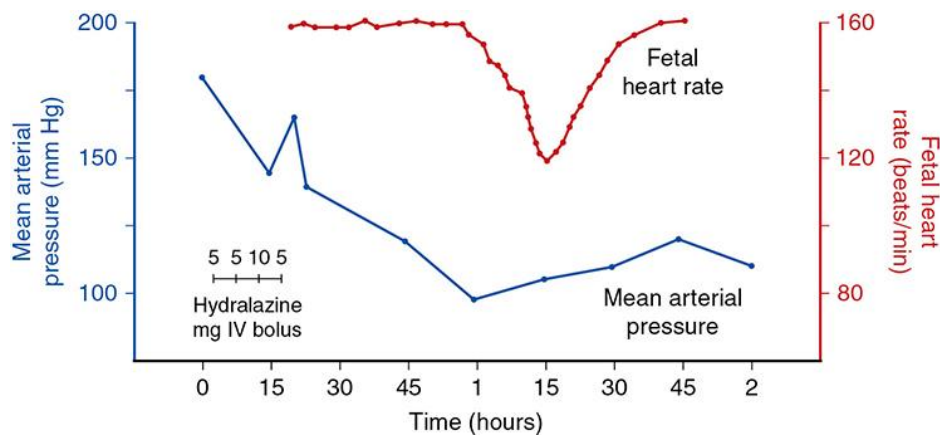


Figure 41-7. Hydralazine given too often: MAP fell from 180 to ~90 mm Hg and the FHR dropped to ~120; fluid raised MAP to ~115 and the fetus recovered.

Diuretics

- Loop diuretics further **compromise placental perfusion** and redistribute an already contracted volume → **not used antepartum to lower BP**; furosemide before delivery **only for pulmonary edema**. May help **postpartum** hypertension.

Fluid therapy

- Crystalloid at **60–125 mL/h** unless unusual losses (vomiting, diarrhea, diaphoresis, hemorrhage).
- ⚠️ Oliguria is common and tempting to treat, but these women already have excess extravascular fluid. Large volumes → **pulmonary and cerebral edema**.
- **Oliguria/anuria:** small incremental boluses to keep urine output **>30 mL/h**. Replace hemorrhage or other losses with gradual boluses. Before neuraxial analgesia, infuse slowly in graded amounts.

Pulmonary edema

- **Most often postpartum.** In eclampsia, first exclude **aspiration** (from convulsion, anesthesia or oversedation).
- Three causes:
 1. **Capillary permeability edema** ($\uparrow$ permeability + $\uparrow$ extravascular oncotic pressure; worsened by vigorous fluids)
 2. **Cardiogenic** (ventricular failure from severe hypertension, more common with **morbid obesity or chronic hypertension**)
 3. A **combination**

Invasive hemodynamic monitoring

- Classic indications: oliguria or pulmonary edema, and "ironically, vigorous treatment of the former causes most of the latter."

- **Task Force: no routine invasive monitoring.** Reserve for severe preeclampsia with **cardiac and/or renal disease**, or **refractory hypertension, oliguria and pulmonary edema**. Noninvasive monitors are unverified.

Plasma volume expansion

- Older studies: serious complications, especially **pulmonary edema**.
- **Amsterdam RCT** (216 women, 24–34 wk): **250 mL 6% hydroxyethyl starch over 4 h twice daily** → **no difference** in outcomes. **Not recommended**.

Analgesia and anesthesia

- **Neuraxial analgesia is ideal:** safe in RCTs. Slow induction with **dilute** agents avoids hypotension and the need for large fluid preloads.
- It **avoids general anesthesia**, where **intubation causes sudden severe hypertension** → pulmonary edema, cerebral edema, ICH. **Airway edema** makes intubation difficult.
- Vigorous crystalloid with epidural → ↑ PCWP; aggressive volume → pulmonary edema, especially **in the first 72 h postpartum**; most **pharyngolaryngeal edema** comes from aggressive fluids.
- General, epidural or CSE are all acceptable with a careful approach. Parkland: a **gentle bolus** with epidural placement.

Blood loss at delivery

- These women **lack pregnancy hypervolemia** and tolerate even normal blood loss poorly.
- ⚠️ **A sharp BP fall soon after delivery usually means excessive blood loss**, not sudden resolution of vasospasm.
- Oliguria after delivery → check **hematocrit frequently**; treat hemorrhage with crystalloid and **blood**.

9. Persistent Severe Postpartum Hypertension

- **8%** of women develop **de novo** postpartum hypertension.
- If IV hydralazine/labetalol or oral nifedipine keeps being needed → **oral maintenance**: labetalol or another β-blocker, **nifedipine XL**, **amlodipine** or another CCB. This **reduces readmission**. SMFM has a postpartum checklist.
- Causes: **mobilized interstitial fluid returning to the circulation**, underlying chronic hypertension, or both.
- **NSAIDs do not aggravate** postpartum hypertension.
- With chronic hypertension + **LVH**, severe postpartum hypertension can cause **cardiac pulmonary edema**.

Furosemide

- RCT (384 women): furosemide → less hypertension at **7 days (6% vs 14%)** and fewer discharged on antihypertensives.
- Nifedipine + furosemide vs nifedipine alone: need for additional agents **8% vs 26%**. **Torsemide**: no benefit.
- **Parkland weight check**: postpartum weight should be **≥10–15 lb below** the last prenatal weight. If it is **higher** (IV fluids with epidural or surgery) **and** severe hypertension persists → **IV furosemide**.

Thrombotic microangiopathy

- Rare atypical persistence of severe preeclampsia–eclampsia after delivery. **Plasma exchange** advocated (e.g., 3 L × 3 = **36–45 donor units**), but the distinction from **TTP/HUS** was unclear.
- Parkland: of **>55,000** women with gestational hypertension (~500,000 pregnancies), **very few** had a TMA.

Reversible cerebral vasoconstriction syndrome (RCVS)

- Persistent hypertension, **"thunderclap" headaches**, seizures and CNS findings. **Diffuse segmental constriction** of cerebral arteries; may cause ischemic or hemorrhagic stroke.
- Triggers include **pregnancy and especially preeclampsia**; more common in women; long-term sequelae uncommon. Best management unclear.

10. Long-Term Consequences

Future pregnancies

- ↑ risk of hypertensive disorders in later pregnancies and later life; hypertension risk may be highest in the **first 6 months** postpartum.
- After preterm preeclampsia, even normotensive later pregnancies have more **preterm birth and FGR**.
- **Earlier onset → higher recurrence:**

Index pregnancy	Recurrence risk
Preeclampsia near term	23%
Nulliparas diagnosed <30 wk	up to 40%

- **HELLP** recurs substantially; even without recurrence, preterm delivery, FGR, abruption and cesarean are ↑.

Cardiovascular morbidity

- Preeclampsia is a **marker for long-term cardiovascular disease** → evaluate hypertensive women in the **first several months postpartum**.
- Pregnancy-attributable hypertension should resolve by **12 wk**; beyond that it is **chronic hypertension**.

Data	Finding
Magpie follow-up (3375, median 26 mo)	20% hypertensive
Meta-analysis	28% hypertension within 2 yr
Danish registry (~15 yr)	Chronic hypertension 5× after gestational hypertension, 4× after mild, 6× after severe preeclampsia
Iceland (gestational hypertension vs controls)	Ischemic heart disease 24% vs 15% ; stroke 9.5% vs 6.5%
Sweden (>400,000, recurrent preeclampsia)	Systolic dysfunction, ↑ ischemic heart disease; also more diastolic dysfunction

- Also **coronary calcification** and **idiopathic cardiomyopathy**. ↑ LV mass index **before** hypertension develops.
- Risk may begin at the woman's **own birth** (as with preterm birth and FGR). Cofactors: **metabolic syndrome, diabetes, obesity, dyslipidemia, atherosclerosis**. ↑ **type 2 diabetes**, later **diabetic retinopathy** and **retinal detachment**.

Table 41-10. Some Long-Term Consequences in Women with Preeclampsia Syndrome

Obstetrical	Cardiovascular	Renal	Neurovascular	Metabolic
Hypertensive disorders	Chronic hypertension	Glomerular dysfunction	Stroke	Type 2 diabetes
Preterm delivery	Ischemic heart disease	Proteinuria	Retinal detachment	Metabolic syndrome
Fetal-growth restriction	Atherosclerosis		Diabetic retinopathy	Dyslipidemia
	Coronary artery calcification		Dementia	Obesity
	Cardiomyopathy		Seizure disorder	
	Thromboembolism		Retinopathy	
	Peripheral vascular disease			

Table 41-11. Long-Term Cardiovascular Morbidities After Recurrent Preeclampsia Compared with a Single Pregnancy with Preeclampsia and a Subsequent Normal Pregnancy

Morbidity (No. of Studies)	Recurrent (No.)	Single (No.)	Risk Ratio ^a
Hypertension (7)	2186	5589	2.33
IHD (3)	428	977	2.40
Heart failure (2)	160	298	2.88
CVA (2)	196	616	1.69
Cardiovascular event (2)	1278	5909	1.57

^aAll *p* values <0.0002. CVA = cerebrovascular accident; IHD = ischemic heart disease.

Renal sequelae

- **~15%** of women with prior preeclampsia have **renal dysfunction** (possibly from preeclamptic AKI).
- Danish study (>1 million): chronic renal disease **6x** after preeclampsia.
- But **15–20%** of preeclamptic women who undergo biopsy **already have chronic renal disease**.
- Long-term: chronic hypertension **55% vs 7%** in controls, with ↑ renovascular resistance and ↓ renal blood flow (cause vs effect unclear).

CNS sequelae

- Eclampsia does have lasting effects: almost all have multifocal perivascular edema, **~¼ cerebral infarction**.
- **Persistent white-matter lesions** on MRI: **40%** after eclampsia vs **17%** of controls; also seen in preeclampsia without seizures. Temporal-lobe white-matter changes and ↓ cortical volume.
- Some studies: subjective **cognitive impairment**, ↑ **dementia**, modest ↑ **seizure disorders**, ↓ vision-related quality of life.

Quick-Recall: Drugs in Severe Preeclampsia–Eclampsia

Drug	Regimen	Key cautions
MgSO₄ IV	4–6 g over 15–20 min, then 2 g/h; continue 24 h postpartum	Therapeutic 4–7 mEq/L ; reflexes lost 10 ; respiratory arrest ≥12 . Obese: may need 3 g/h . Reduce maintenance only if creatinine ↑
MgSO₄ IM (Pritchard)	4 g IV (20%) + 10 g IM (50%, 5 g each buttock), then 5 g IM q4h	Only if patellar reflex present, respirations normal, urine >100 mL/4 h
Recurrent seizure	2 g of 20% MgSO₄ IV slowly (×1–2)	Refractory: small dose midazolam lorazepam; avoid prolonged use
Calcium gluconate	1 g IV + stop MgSO ₄	For Mg respiratory depression; intubate if arrest
Hydralazine	5 mg IV/IM , then 10 mg q15–20 min (≤3 doses)	Tachycardia, headache; overshoot → fetal distress
Labetalol	10 → 20 → 40 → 40 → 80 mg IV q10 min (or 1–2 mg/min)	Asthma , bradycardia, hypotension
Nifedipine IR	10–20 mg orally , repeat after 20 min	Never sublingual ; tachycardia, headache
Furosemide	IV, postpartum	Antepartum only for pulmonary edema
BP target	SBP ≤160 , DBP ≤110	Don't drop the diastolic too low (placental perfusion)
IV fluids	60–125 mL/h crystalloid	Oliguria → small boluses for >30 mL/h ; no volume expansion

Key Numbers to Memorize

- Admit if new BP **≥140/90**; BP checks **every 4 h**
- Deliver nonsevere preeclampsia at **≥37 wk** (Parkland **38 wk**); testing **twice weekly** (nonsevere) vs **daily** (severe)
- HYPITAT-II: maternal adverse outcome **1.1% vs 3.1%**, RDS **5.7% vs 1.7%**
- Parkland day unit: **~90%** have recurrent hypertension
- Expectant management of severe disease: gains **~1 wk** (5–12 days); PMR **~90/1000**; Memphis 18–27 wk PMR **87%**
- Survival: **0% <23 wk**, **18% at 23 wk**, **~60% at 24–26 wk**, **~90% at 26 wk**
- Delivery triggers <34 wk: steroids first dose only for maternal indications; **48-h delay** for PPROM, FGR, oligohydramnios, **reversed EDF**, worsening renal function
- Eclampsia: abruption **10%**, maternal death **1%**; FHR recovers in **2–10 min**; coma **≤5%**; blindness **~10%**; recurrent seizure on MgSO₄ **≤15%**
- Magpie: **58%** ↓ eclampsia; nonsevere untreated → **1 in 100** seizures
- MgSO₄ therapeutic **4–7 mEq/L**; reflexes lost **10**; respiratory arrest **≥12**; Ca gluconate **1 g**
- Postpartum de novo hypertension **8%**; postpartum weight should fall **≥10–15 lb**
- Recurrence **23%** (term) to **40%** (<30 wk); hypertension **20%** at 26 mo; chronic hypertension **4–6×** at 15 yr
- Renal dysfunction **~15%**; white-matter lesions **40% vs 17%**